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Prompting for Few-shot Adverse Drug Reaction Recognition from Online Reviews

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***Abstract*—The growing popularity of social network sites (SNS) contributes to the abundance of healthcare-related user-generated content (UGC), which can be used for identifying adverse drug reactions (ADR). Most previous works on automatic ADR detection, however, require a large amount of labeled data to reach acceptable recognition performance, deemed labor-intensive and inefficient when building an ADR detection model for UGC. To address this issue, this study proposes to apply LightNER [1], a few-shot Named Entity Recognition (NER) approach that exploits the power of a pre-trained language model (i.e., BART) and introduces a small set of prompt parameters to enable few-shot learning. Our approach first warms up the prompt parameters for learning knowledge for NER from rich-resource datasets. Then, the approach only requires a few labeled UGC to fine-tune the parameters for satisfactory ADR recognition performance. In our experiment, we collect a review dataset for a migraine drug called MigraineReviews and adopt three rich-resource datasets, including one general-purpose dataset and two datasets for clinical notes. The experimental results demonstrate that BART with prompt parameters helps transfer knowledge across datasets even if they are of different domains, thus reducing the need for a large amount of labeled ADR data. Our approach performs competitively with a limited size of labeled data (e.g., 5-shots).**

***Index Terms*—adverse drug reaction, online review, few-shot learning, prompt learning, named-entity recognition**

I. Introduction

An adverse drug reaction (ADR) is a harmful or unpleasant reaction resulting from the use of a medicinal product [2]. Surveys show that ADRs contribute to 5% of all hospital admissions and occur in 10-20% of inpatients [3]–[5]. Although every legitimate drug has been verified for its safety and effectiveness by years of clinical trials, it hardly finds all possible ADRs due to the limited trial samples and test period. Often, some unexpected ADRs can be found after a drug is placed on the market. Once we thoroughly know ADRs of a drug, the physician may make a better prescription for a patient to meet his (her) conditions. For example, if a drug has a higher odd causing ADRs to middle-aged women than other groups, physician should prevent prescribing that drug to them. To monitor the safety of post-market drugs, spontaneous report system is the de facto method to collect ADRs, yet it takes time to collect enough evidence and has under-reporting issues [6].

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Thanks to the development of social networking sites (SNS), people can share personal experiences of diseases, report reflections of medicines, and communicate with other medical professionals or patients. SNS has gradually become the main channel for people to exchange health-related experiences. SNS complements spontaneous report systems by collecting more feedback from a wider variety of patients in a more efficient way than ever before. Researchers can gather tons of user-generated contents (UGCs) from SNS and propose methods for recognizing ADRs from texts [7]–[11].

Previous approaches on automatic ADR recognition can be grouped into semi-supervised approach and supervised approach. Most of semi-supervised approaches will first identify potential ADRs using existing lexical resources such as Unified Medical Language System (UMLS). Once potential ADRs are identified, association rule [12] or topic model [7], [11] can be applied for calculating the relatedness between potential ADRs and drugs. Since the vocabulary from pre-compiled lexical resources may not cover well on health terms used by laypeople, the low recall is a critical concern [13], [14]. On the other hand, supervised approaches can be realized for solving binary classification problem [8]–[10], [15] or sequence labeling problem [9], [16]–[19]. However, both problems require sufficient volume of data to train the model. It is time-consuming and labor intensive to collect large-scale annotations for UGCs.

The shortage of annotated data for UGCs necessitates the development of few-shot ADR recognition models. Due to the current development of prompt learning, using pre-trained language models (PLM) for few-shot settings has shown efficacy and efficiency [20], [21]. Prompt learning allows us to tune a smaller set of parameters (i.e., prompt) without fine-tuning the

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whole parameters in PLM, thereby requiring only a few data to adapt the PLM for a given task. For adopting prompt learning for ADR recognition, we rely on LightNER [1] that incorporates prompt parameters with BART [22] for NER tasks. The prompt parameters of LightNER are designed to learn task-relevant knowledge from rich-resource dataset D^{rich} and transfer the knowledge to low-resource dataset D^{low}. In this study, we have three Named Entity Recognition (NER) annotated datasets: CoNLL2003 (news), BC5CDR (biomedical literature), and N2C2(clinical reports) as our D^{rich}. We collect MigraineComments as D^{low} that are comments from WebMD discussing ADRs for a drug of migraine. With the LightNER and its prompt parameters, we conjecture that *even though D^{rich} (e.g., news, biomedical literature, and clinical reports) and D^{low} (i.e., ADR recognition for online drug reviews) are of different domains, the knowledge of sequence labeling task can be transferred across dataset, thereby facilitating few-shot learning for D^{low}*. We propose a transfer strategy for LightNER to verify this conjecture. As a result, we reach the following conclusions from our experiments:

1) By fine-tuning pre-trained language model (BART [22]) with few-shot data, it performs better in ADR recognition than knowledge base-driven detection approach (sciSpacy [23]).
2) By introducing a small set of learnable prompt parameter with BART [22], we can transfer NER task knowledge from rich-resource dataset and help recognize ADR in low-resource dataset.
3) By warming up prompt parameters using rich-resource dataset, we can further fine-tune the prompt parameters using low-resource data to reach a competent ADR recognition performance for low-resource dataset.

II. Dataset

We collect the MigraineReviews dataset from WebMD, one of the most popular SNSs for healthcare. We refer to the extensive drug database on WebMD and collect 194 reviews for Topamax, which is a drug for treating migraine. The reviews varied in length, ranging from 1 to 254 words, with an average of 48.15 words. Two authors manually read each review and annotate the following 6 entity types: Drug, ADR[1], Symptom, Dose, Frequency, and Disease. Fig. 1 shows an annotation example for a review. Table I presents the total number of entities for each entity type and the number of sentences containing some entities for each entity type. Table II reports the inter-annotator agreement using F-measure, in which we treat the annotations from a annotator as ground truth and compare them with those from the other annotator [24], [25]. The average F-measure is 0.81, indicating the high agreement between two annotators. Both authors discuss the disagreements to reconcile all of them.
We prepare three rich-resource datasets:

[1]Most ADRs are for Topamax, and very few ADRs (n=12, 3%) are for other drugs. We remain all ADRs because they are all helpful for training an ADR recognition system.

I have been taking 50 mg dose 2x a day frequency since 11/30/18. The first week a migraine Symptom put me in the ER. I am still suffering with the usual cluster headache Disease and not finding much relief. I am not sure if this new drug is working or if I should continue trying it. I do like the loss of appetite ADR that comes with taking tomamax Drug because everything else I have tried causes weight gain ADR and I do not need that! Not sure what to do at this point. Suffer with the headaches ADR and foggy brain ADR or ask to try another medication and go through those side effects....

Fig. 1. Example of review annotation

TABLE I
Entity statistics for MigraineReviews

	ADR	Disease	Drug	Symptom	Dose	Frequency
#counts	380	158	107	59	60	47
#sentences	132	113	64	46	33	24

1) CoNLL2003 [26] is a classic Named Entity Recognition (NER) task dataset. CoNLL2003 has four entity types, namely PERSON, LOCATION, ORGANIZATION, and MISC (i.e., entities that do not belong to the above three classes). It contains Reuters news articles, with 6,600, 7,140, 6,321, and 3,438 annotations on PERSON, LOCATION, ORGANIZATION, and MISC, respectively.
2) BC5CDR [27] contains 1500 abstracts of biomedical literature from the PubMed website. BC5CDR contains 5,203 chemical annotations and 4,182 disease annotations.
3) N2C2 [28] is the dataset provided by the 2018 National NLP Clinical Challenges. This dataset consists of drug mentions and relations in clinical notes. We only consider eight entity types, including 837 adverse drug event annotations, 4,484 strength annotations, 3,378 dosage annotations, 482 duration annotations, 4,482 frequency annotations, 5,427 form annotations, 4,305 route annotations, and 4,127 reason annotations.

CONLL2003 is a completely different domain compared with MigraineReviews. Both BC5CDR and N2C2 are in the health-related domain, but their contexts are in biomedical literature and clinical reports, which are somewhat different from reviews used in SNS. Nevertheless, they are all designed NER sequence labeling task.

III. Methodology

To transfer knowledge from rich-resource dataset (i.e., CoNLL2003, BC5CDR, and N2C2) to low-resource dataset (i.e., MigraineReviews), we adopt LightNER [1] as our sequence labeling model. In the following subsections, we will describe the architecture of LightNER and the approach to transfer task knowledge across dataset.

A. LightNER

Fig. 2 illustrates the architecture of LightNER [1], which is a seq2seq architecture built on pre-trained BART model [22]. BART adopts a standard sequence-to-sequence Transformer architecture [29], comprising an encoder and a decoder.

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TABLE II
INTER-ANNOTATOR AGREEMENT FOR EACH ENTITY TYPE

	Precision	Recall	F-measure	Support
ADR	0.78	0.79	0.79	380
Disease	0.90	0.85	0.88	158
Drug	0.88	0.90	0.89	107
Symptom	0.71	0.69	0.70	59
Dose	0.94	0.78	0.85	60
Frequency	0.74	0.60	0.65	47
Weighted AVG	**0.82**	**0.80**	**0.81**	**811**

Specifically, BART's encoder is bidirectional and allows for capturing contextual information from texts, and BART's decoder is a autoregressive decoder which excels in text generation task.

As we consider ADR recognition as a NER task, we formulate the problem as follows. The input sequence is $X = \{x_1, ..., x_n\}$ with n tokens and the output sequence $Y = \{y_1 = (e_1^{start}, e_1^{end}, g_1), ..., y_i = (e_i^{start}, e_i^{end}, g_i)\}$ is a set of entity spans and their types, where e^{start} (e^{end}) represent start (end) token index of an entity. $g_i \in G$ is a entity type. Given a input sequence X and its entity sequence Y, LightNER aims to model $P(Y|X)$.

B. Encoder

For dealing with such a NER task, LightNER implements the pointer mechanism [30], in which the encoder encodes the input sequence X into embeddings $\mathbf{H}^e \in \mathbb{R}^{n \times d}$ by

$$\mathbf{H}^e = \text{Encoder}(X), \tag{1}$$

where n is number of tokens, and d is the embedding dimension.

C. Decoder

To output entities and their corresponding entity types, the decoder will return an index either in the input sequence *X* or in the set of entity types *G* at each time step t. The decoder requires the output states from encoder $\mathbf{H}^e$ and the prediction before t, $\hat{Y}_{<t}$ for generating the embedding at time t, defined as follows[2]:

$$\mathbf{h}_t^d = \text{Decoder}(\mathbf{H}^e, \hat{Y}_{<t}) \tag{2}$$

Once $\mathbf{h}_t^d \in \mathbb{R}^d$ is computed, the index probability distribution P_t can be acquired by

$$\mathbf{E}^e = \text{TokenEmbedding}(X), \tag{3}$$

$$\hat{\mathbf{H}}^e = \alpha \times \mathbf{H}^e + (1 - \alpha) \times \mathbf{E}^e, \tag{4}$$

$$\mathbf{G}^d = \text{TokenEmbedding}(G), \tag{5}$$

$$P_t = \text{Softmax}([\hat{\mathbf{H}}^e \otimes \mathbf{h}_t^d; \mathbf{G}^d \otimes \mathbf{h}_t^d]), \tag{6}$$

[2]Note that the decoder returns index, so every index is transformed to token before feeding into decoder. Please refer to [30] for more details.

where $\mathbf{E}^e, \hat{\mathbf{H}}^e \in \mathbb{R}^{n \times d}$; $\mathbf{G}^d \in \mathbb{R}^{|G| \times d}$; $P_t \in \mathbb{R}^{(n+|G|)}$; $\otimes$ means dot product; $[\cdot;\cdot]$ means concatenation in the row direction. By $\hat{\mathbf{H}}^e \otimes \mathbf{h}_t^d$, we obtain the logits of words, in which the word with higher logit has higher opportunity to be drawn. Similarly, $\mathbf{G}^d \otimes \mathbf{h}_t^d$ predicts the logits of possible entity types. The embedding of each entity type $g^d \in G^d$ is obtained by querying token embedding layer. When entity type contains multiple tokens such as <Adverse_Drug_Reaction>, we learn weights for averaging token embeddings: $\beta_1 \times$ TokenEmbedding(Adverse) + $\beta_2 \times$ TokenEmbedding(Drug) + $\beta_3 \times$ TokenEmbedding(Reaction). As a result, either a word index or an entity type index at time t can be drawn from P_t, which is illustrated as word probability and tag probability as shown in Fig. 2. The index can be translated into a token or an entity type $\hat{y}_t$. Take Fig. 2 as example, for an input sequence X = "I do like loss my appetite", suppose the decoder has already generated $\hat{Y}_{<t}$ = "loss", "my", and "appetite". The most probable prediction for next step t would be an index to "<Adverse_Drug_Reaction>". For the training phase, we can use the negative log-likelihood loss based on the collected decoded sequence $\hat{Y}$.

D. Guidance Module

In order to learn the task knowledge across dataset, LightNER introduces a set of learnable prompt parameters by modifying the attention function in the transformer architecture, which is shown in Fig. 3. The attention function provides the weights of the tokens in the input sequence for guiding decoder. To compute the attention weights for the input representation at layer l, $X^l \in \mathbb{R}^{n \times d}$, it requires query matrix Q^l, key matrix K^l, and value matrix V^l:

$$Q^l = X^l W^Q, K^l = X^l W^K, V^l = X^l W^V, \tag{7}$$

where $W^Q, W^K, W^V \in \mathbb{R}^{d \times d}$ are frozen parameters loaded from pre-trained BART. For adapting attention function to the NER task, two prompt parameters $\phi_K^l, \phi_V^l \in \mathbb{R}^{P \times d}$ are concatenated to key matrix K^l and value matrix V^l, respectively. Since prompt parameters are learnable, they provide the room and flexibility to carry on the task knowledge across datasets. As a result, the updated attention function is calculated as follows:

$$\text{attention}^l = \text{softmax}(\frac{Q^l[\phi_K^l; K^l]^T}{\sqrt{d}})[\phi_V^l; V^l] \tag{8}$$

LightNER is the stack of L transformer layers, where each layer adopts the transformer structure as presented in Fig. 3. Therefore, the prompt parameters in all L layers will contribute to the calculation of encoder's output $\mathbf{H}^e$ (Eq. 1) and decoder's output $\mathbf{h}_t^d$ (Eq. 2).

E. Transfer Strategy

We adopt a strategy called Fixed-LM Prompt Tuning [31]. As shown in Fig. 2, we fix all LM parameters and keep them unchanged. The learnable parameters of the guidance module are prompt parameters: $(\Phi_K, \Phi_V) = \{(\phi_K^l, \phi_V^l) : l \geq 1, l \leq$

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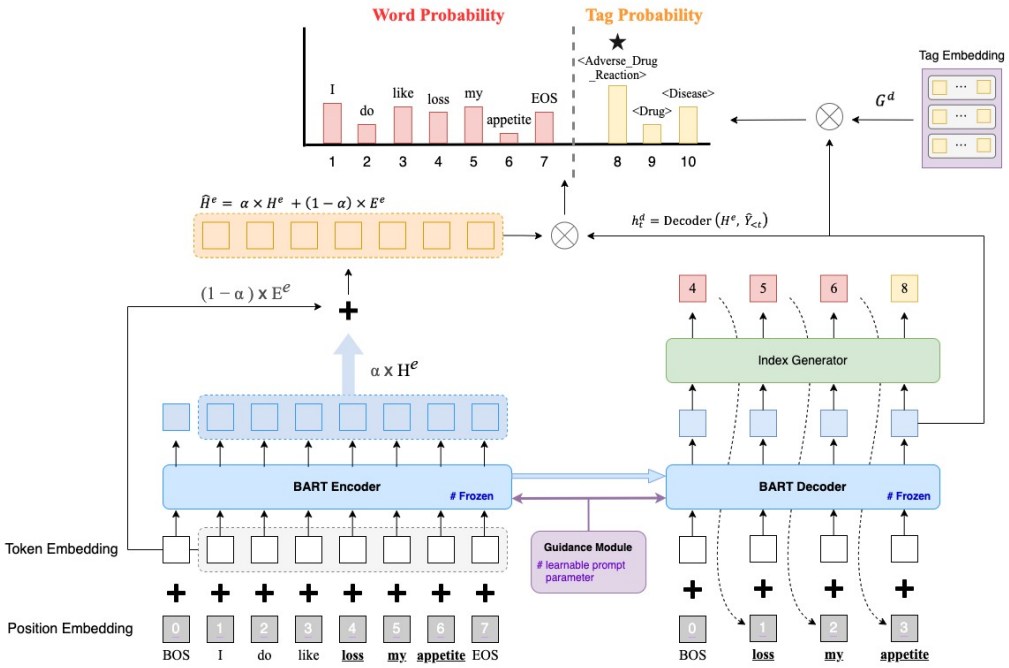


Fig. 2. The architecture of LightNER

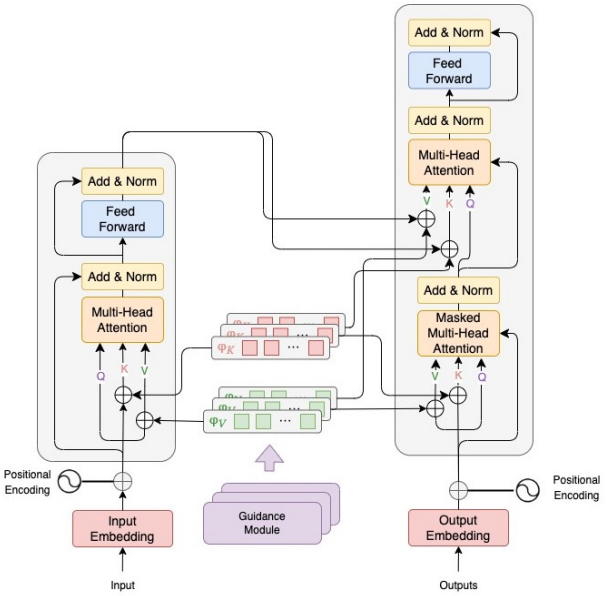


Fig. 3. A seq2seq architecture with the updated transformer layer; guidance module is realized by introducing prompt parameters in attention calculation.

$L\}$, providing the room for learning knowledge of sequence labeling task. Algorithm 1 presents the two main steps for transferring knowledge from D^{rich} to D^{low}:

1) Warm-up step (line 1-2): given a rich-resource dataset, we first initiate β^{rich} if there are any multi-token entity types in D^{rich}. We then tune prompt parameters (Φ_K, Φ_V) and β^{rich} using all labeled data $(\mathbf{X}^{rich}, \mathbf{Y}^{rich}) \in D^{rich}$.
2) Fine-tune step (line 3-6): the tuned prompt parameters obtained from previous step become the initial states for further fine-tuning prompt parameters. Instead of using all labeled data from D^{low}, we shuffle the data in D^{low} and randomly select n-shot data $\{(\mathbf{X}_i^{low}, \mathbf{Y}_i^{low}) : i \geq 1, i \leq n\}$ as labeled data. Note that when the entity types are different between D^{rich} and D^{low}, β^{low} should be initialized since β^{rich} only works for entity types of D^{rich}. After fine-tuning, we can use LightNER, equipped with updated prompt parameters (Φ_K, Φ_V) and β^{low}, and follow Eq. 1 to Eq. 6 to predict entities and their entity types for any token sequence $X \in \mathbf{X}^{low}$.

IV. Experiment

A. Experimental Settings

1) Model Settings: We adopt pre-trained BART-large [22] as our PLM for LightNER, which has 12 transformer layers for both encoder and decoder. We add prompt parameters at each layer and set $|P| = 10$. At warm up step, we adopt the batch

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Algorithm 1 Transfer Strategy

Require: LightNER, D^{rich}, D^{low}, n

1: initialize β^{rich} for multi-token entity types of D^{rich}
2: update (Φ_K, Φ_V) and β^{rich} using $(\mathbf{X}^{rich}, \mathbf{Y}^{rich}) \in D^{rich}$
3: shuffle the data in D^{low}
4: select n-shot data $\{(\mathbf{X}_i^{low}, \mathbf{Y}_i^{low}) : i \geq 1, i \leq n\} \in D^{low}$
5: initialize β^{low} for multi-token entity types of D^{low}
6: update (Φ_K, Φ_V) and β^{low} using n-shot labeled data
7: **return** updated LightNER

size as 16 and learning rates as $2e-5$. We train the model for 30 epochs. At fine-tune step, we use the batch size as 8 and learning rates as $2e-5$. We fine-tune the model for 100 epochs and evaluate after 90 epoch. All hyper-parameters are picked up by monitoring the loss on validation set. We keep using the same optimizer (ADAM) as adopted by BART-large.

2) Evaluation Approach: We separately use each rich-resource dataset (CoNLL2003, BC5CDR, and N2C2) for warm-up. Then, we fine-tune the model using low-resource dataset and evaluate the ADR recognition performance on low-resource dataset (MigraineReviews). Since we aim to study the transfer ability of LightNER, we only fine-tune LightNER by a small set of data (n-shot) from MigraineReviews. We sample $n \in \{5, 10, 15, 20\}$ sentences for each entity type from MigraineReviews. Take ADR entity as example, as $n = 5$, we sample 5 sentences out of 132 sentences containing ADRs (cf. Table I). Each experiment repeats line 3-7 of Algorithm 1 5 times for selecting different set of sentences for obtaining different fine-tuned models, and the average performance is reported for robust results.

3) Compared Methods: We compare LightNER with two methods. The first one is **sciSpacy** [23], which is a recent NLP tookit for biomedical text. We adopt "en_core_sci_sm" model and resolve the abbreviation. **sciSpacy** detects medical entities by consulting a medical knowledge base, UMLS. The other method is called **NoTransfer**, in which we skip the warm-up step in transfer strategy and directly fine-tune LightNER using MigraineReviews.

4) Metrics: Although we have 6 entity types as shown in Table I, we only report the performance for ADR to meet the goal of this study. The performance metrics include precision, recall, and F1 score. Almost all ADRs in our MigraineReview dataset refer to Topamax (a migraine medication), so we skip relation extraction in our study.

B. Results

Table III presents the performance of **NoTransfer**, where performance increases with the number of shots. Note that for **NoTransfer** we directly fine-tune LightNER using MigraineReviews without using any rich-resource dataset for warm-up. Compared to the performance of **sciSpacy** (Precision = 59.46%, Recall = 37.09%, F1-score = 45.68%), **NoTransfer** starts to outperform **sciSpacy** after 15-shots. This experiment shows that with appropriate amount of data, simply fine-tuning prompt parameters on PLM achieves better performance than that of **sciSpacy**, which is based on knowledge based-driven recognition model. The results also imply that pre-trained BART-large contain large general knowledge, and it is possible to fine-tune prompt parameters using few-shot data for ADR recognition.

TABLE III
THE RECOGNITION PERFORMANCE OF **NOTRANSFER**

	5-shots	10-shots	15-shots	20-shots
Precision	40.28	57.57	61.63	63.11
Recall	19.93	33.89	39.66	45.47
F1-score	26.22	42.52	48.11	52.77

Table IV presents the performance of LightNER that are first warmed up using rich-resource dataset. Likewise, performance increases when number of shots grows. Comparing Table III and Table IV, we know that warming LightNER up by any rich-resource dataset contributes to the increase in performance. With LightNER, it only need 5-shots from MigraineReviews for fine-tuning to well outperform sciSpacy and surpass the performance of **NoTransfer** with 20-shots. This observation demonstrates that LightNER can transfer task knowledge regardless of the domain of dataset. LightNER gets the highest performance, though only by a small margin, when it warms up by N2C2, because N2C2 shares the same domain and has similar entity types with MigraineReviews. To test the ability of zero-shot learning, we further test 1-shot for LightNER warmed up by N2C2; however, the precision, recall, and F1-score are 59.94, 29.31, and 39.36, respectively. The fall in recall and F1-score demonstrates that the current approach for zero-shot learning obtains suboptimal result.

TABLE IV
THE RECOGNITION PERFORMANCE OF TUNED LIGHTNER

Dataset	Metric	5-shots	10-shots	15-shots	20-shots
CoNLL03	Precision	57.96	63.62	67.92	66.80
	Recall	55.92	59.97	61.59	66.61
	F1-score	56.83	61.64	64.58	66.60
BC5CDR	Precision	65.88	66.85	69.42	66.34
	Recall	58.88	62.39	65.37	69.69
	F1-score	62.18	64.46	67.29	67.90
N2C2	Precision	65.66	67.06	67.41	68.18
	Recall	57.81	63.33	67.61	70.23
	F1-score	61.42	65.10	67.51	69.09

Overall, the results support our conjecture mentioned in introduction that we can use LightNER to transfer task knowledge from rich-resource dataset to low-resource dataset even when they are of different domains. With transferred knowledge, it only require few-shot data for fine-tuning in low-resource dataset to achieve acceptable performance.

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C. Error Analysis

We conduct error analysis for the prediction error of LightNER. We define three types of prediction results. Type I result refers to the case when the ADR prediction correctly match the ground truth; this is a correct case. Type II result occurs when LightNER falsely predicts non-ADR string as ADR type. Type III result happens when LightNER fails to identify the ADR string annotated in the ground truth. Table V presents the counts of prediction types of LightNER (N2C2), where we separate ADRs into phrases (e.g., loss of appetite, weight gain, and foggy brain) and single words (e.g., headache, sleepless and nausea). As can be seen, most results are attributed to Type I, showing the capability of LightNER for recognizing ADRs regardless of the form of ADR (i.e., phrase or single word). LightNER has more Type II errors than those of Type III. This suggests LightNER is a bit sensitive and it sometimes predicts ADR aggressively.

TABLE V
THE COUNTS OF PREDICTION TYPES OF LIGHTNER (N2C2)

	Type I	Type II	Type III
Phrase	75	17	7
Single word	51	10	2
Total	126	27	9

We provide the confusion matrix for analyzing the prediction error across different entity types. Fig. 4 shows confusion matrixes for **NoTransfer** and LightNER (N2C2), where each row presents the instances of each actual class while each column presents the instances of each predicted class. In **NoTransfer**, it has lower F1 score on ADR since it cannot rely on the prompt parameters warmed up from rich-resource dataset. We observe that many Type III errors exist, as this version of LightNER fails to identify entities for ADR (79 errors), Disease (15 erroes), Drug (17 errors), and Symptom (12 errors). When LightNER is warmed up using N2C2, Type III errors are greatly reduced. For example, only 9 ADR errors, 1 Disease error, 1 symptoms error remain. However, Type II error become the main issues (e.g., 27 ADR errors), and this observation provides a direction for future studies to improve the recognition performance.

V. RELATED WORKS

Automatic ADR recognition consists of entity extraction and relation extraction [12]. Existing approaches to automatic ADR recognition can be categorized into semi-supervised and supervised approaches. For semi-supervised approaches, lexicon- and knowledge-based resources are the primary databases for entity extraction. To identify relations, the association rule algorithm [12] and topic model [7], [11] have shown efficacy for identifying ADRs. However, semi-supervised approaches heavily rely on expert-compiled lexica for entity extraction. Most lexica are for professional intent, which is unsuitable for non-professional data such as UGC. The terminology gap between professionals and the general public results in terrible alignment performance [13], [14]. For example, people would use “cannot sleep” and “throw up” when describing symptoms on social media rather than use formal terms like “insomnia” and “vomit.”

Supervised approaches have also been widely used in discovering ADRs. The binary classification problem and sequence labeling problem are two common realizations. The binary classification problem is to identify if the textual input discusses ADR [8], [10], [15], [32]. Nevertheless, determining whether text content relates to ADR cannot provide enough insights. In most cases, people prefer to know which text span of a given text is ADR, resulting in the sequence labeling problem that aims to identify potential ADRs from a text. Most studies [16]–[18] adopt a pipeline-based approach that separately learns entity and relation extraction model. This pipeline-based approach is prone to error propagation and fails to capture cross-task knowledge between models. Joint-based approach [19], [33] is introduced for performing entity and relation extraction simultaneously. For example, MTTLADE [19] implements a multi-Task transfer learning-based method based on sciBERT, which performs better than pipeline-based approaches on two clinical datasets. The main drawback of the supervised approach is the requirement of sufficient labeled data, which impedes its applicability in UGCs. The UGCs, such as reviews on drug feedback, are of high volume. The contexts (symptoms and adverse drug effects) will differ according to target diseases and medicines. It is labor-intensive and time-consuming to annotate UGCs. Therefore, this study aims to provide an approach that requires a small amount of label data (few-shot) while having an acceptable recognition performance.

VI. CONCLUSION

A significant number of online drug reviews from SNS provides a great opportunity to reveal a drug’s undiscovered ADRs. This study proposes a few-shot ADR recognition approach based on LightNER, which expands pre-trained BART by a set of learnable prompt parameters. The experiments demonstrate that task knowledge can transfer from a rich-resource dataset (i.e., CoNLL03, BC5CDR, and N2C2) to a low-resource dataset (i.e., MigraineReviews), accordingly facilitating few-shot ADR recognition in a low-resource dataset (i.e., MigraineReviews). Our proposed approach is capable of prompting the knowledge pre-learned from PLM. More importantly, the approach only requires a few labeled data, making it easily adapt to online reviews of different drugs. In addition, the approach fine-tunes a relatively small set of parameters (i.e., prompt parameter) for ADR recognition task, which speeds up the development of recognition model. For practitioners, our lightweight approach helps efficiently monitor the safety of pharmaceutical drugs that facilitates drug safety surveillance and benefits multiple stakeholders such as patients, pharmaceutical companies, physicians, and healthcare providers.

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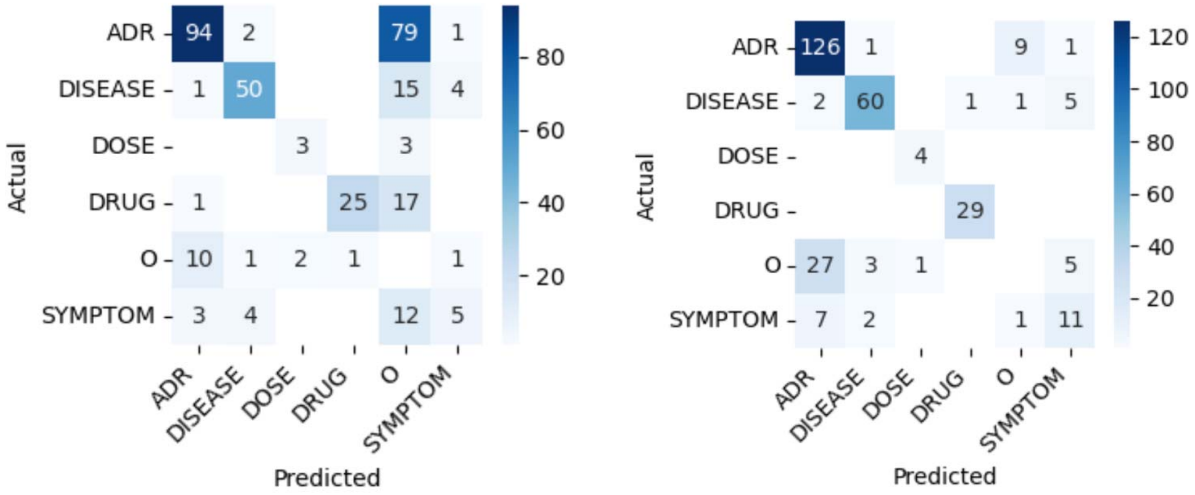


Fig. 4. The left figure is the confusion matrix of **NoTransfer**, and right figure is the confusion matrix of LightNER (N2C2). Both adopt 20-shots for fine-tuning.

The way that we collect drug comments from WebMD eases our efforts to implement a relation extraction model because most of the identified ADRs belong to a specific drug (i.e., Topamax), as discussed in Section II. Therefore, this paper focuses on proposing a few-shot approach for ADRs. The relation extraction is needed when one collects drug-related comments from other platforms, such as Twitter and Reddit. When a comment discusses ADRs for multiple drugs, our proposed approach merely detects entities such as ADRs and drugs. Finding a few-shot (or even zero-shot) approach for extracting the relation between medical entities (e.g., ADR and drug) remains a research question for future studies.

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